

VENKATA HARSHITHA BATHALA

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PROFESSIONAL SUMMARY

Software Engineer with experience in full-stack web development, mobile application development, and backend systems. Proficient in Java, Python, C++, and Dart, with expertise in REST API design, cloud services (AWS, Azure, Firebase), and database management (SQL, MongoDB). Skilled in Agile methodologies, CI/CD automation, and test-driven development, with proven ability to deliver scalable, user-centric solutions across web, mobile, and distributed environments.

EDUCATION

- University of Texas at Arlington** | Master of Sciences in Computer Science – Texas CGPA: 3.9
- Annamacharya Institute of Technology & Sciences** | Computer Science & Engineering - India CGPA: 8.34

EXPERIENCE

Software Engineer | University of Texas at Arlington Jul 2024 – May 2025

- Developed comprehensive **web portal** for **200+ CSE Faculty** using HTML5, CSS3, JavaScript, PHP
- Implemented LAMP stack with Apache, MySQL, secure authentication, and dynamic content delivery
- Consolidated UTA portal information of **500+ inventory items**, policies, forms, and faculty requests
- Configured **20+ lab systems**, implemented LUKS encryption and Linux security protocols
- Created **15+ automated Bash scripts** for system maintenance, user management, and access control
- Optimized virtualization resources and monitored **2 UTA servers** for resource usage and **99%+ uptime**
- Worked with Debian-based (Ubuntu) and RPM-based (CentOS, Red Hat, Fedora) systems

Associate Professional Software Engineer | DXC Technology, Chennai Jan 2023 – Dec 2023

- Developed **scalable full-stack applications** using **React.js** and **Node.js (backend)**, improving feature delivery speed by 30%.
- Designed and optimized **RESTful APIs** to integrate enterprise applications with backend services for high-volume data aggregation.
- Built and maintained **SQL schemas** to improve query execution speed by 25% and reduce backend processing time.
- Streamlined delivery with **CI/CD pipelines (Jenkins, Docker, Git)** enabling zero-downtime automated deployments.
- Enhanced system reliability with monitoring and alerting solutions, reducing incident response time by 20%.
- Deployed and optimized applications on **AWS** with Linux-based infrastructure, leveraging **API Gateway, Lambda, and Load Balancers** for maximum scalability and uptime.

PROJECTS

Gambol Spark – Sports Retail & Reservation Mobile App [Git Repository](#)

- Developed a Android application using **Flutter & Dart**, integrating sports accessories retail with field reservation features.
- Implemented **Firebase backend services** for authentication, real-time database, and secure cloud storage to enhance user experience.
- Designed and optimized **user-friendly UI/UX** in VS Code, ensuring seamless navigation and flexible booking options for customers.
- Gambol Spark Application – [Preview Link](#)

Distributed Systems Fault Tolerance [Git Repository](#)

- Implemented **SWIM** protocol for failure detection using gRPC, Python, Docker containers
- Engineered **Two-Phase Commit (2PC)** system with Protocol Buffers for transaction coordination
- Developed **Raft consensus** with leader election and log replication using Docker orchestration
- Created fault injection testing framework simulating network partitions and node failures

Smart HealthCare Hub [Git Repository](#)

- Built **full-stack healthcare platform** for patients, providers, administrators, pharmacists user types with secure **RBAC**.
- Developed responsive **React front-end** and **PHP/MySQL** backend with Axios API communication
- Implemented encryption, secure authentication, role-based access control for healthcare compliance
- Designed scalable database schema with normalized structures and query optimization

Assistive Visual Question Answering System [Git Repository](#)

- Developed a multimodal AI system to assist visually impaired users by integrating computer vision and NLP for image understanding and question answering.
- Achieved **71.7%** Top-1 and **98.4%** Top-5 accuracy using PaliGemma-2 embeddings classification model
- Implemented Vision Transformers (CLIP, SigLIP, ViT) with GPT integration reaching **65%** semantic accuracy
- Preprocessed VizWiz dataset (31,000 image-question pairs) using transfer learning and multimodal alignment

TECHNICAL SKILLS

Languages: Python, Java, C/C++, Go

Frontend Development: React, TypeScript, JavaScript, HTML5, CSS3, SCSS

Backend & APIs: Node.js, Django, Flask, Spring Boot, RESTful APIs, Microservices, API Design, gRPC

Cloud: AWS (EC2, S3, ECS, Lambda, DynamoDB, RDS, Step Functions), Azure (VMs, Load Balancer, VMSS, NSGs)

Databases: MySQL, MongoDB, MariaDB, DynamoDB, Firebase, Schema Design, Query Optimization, Data Modeling

Machine Learning & AI: PyTorch, TensorFlow, Keras, Scikit-learn, NLTK, Hugging Face, BERT, Computer Vision, NLP

DevOps: Git, Jenkins, Docker, Kubernetes, Ansible, Bash scripting

Tools: JIRA, Apache, Tomcat, Excel, Power BI, Tableau

OS: Linux, Unix, Windows

CERTIFICATIONS

- AWS Certified Cloud Practitioner
- Microsoft Certified: Azure Fundamentals (AZ-900)
- AIGPE Lean Six Sigma Yellow Belt